

PN DEVICES

DIODE



C : E, (B)

SANDWICHED

18.4 TRANSISTORS

A transistor consists of a single crystal of germanium or silicon which is grown in such a way that it has three regions (Figs. 18.14 & 18.15).

In Fig. 18.14 the central region is p type which is sandwiched between two n type regions. It is known as n-p-n transistor. In Fig. 18.15, the n type central region is sandwiched between two p type regions. It forms a p-n-p transistor. The central region is known as base and the other two regions are called emitter and collector. Usually the base is very thin, of the order of 10^{-6} m. The emitter and collector have greater concentration of impurity. The collector is comparatively larger than the emitter. The emitter has greater concentration of impurity as compared to the collector.

It can be seen in Figs. 18.14 and 18.15 that a transistor is a combination of two back to back p-n junctions: emitter-base junction and collector-base junction.

For normal operation of the transistor, batteries V_{BB} and V_{CC} are connected in such a way that its emitter-base junction is forward biased and its collector base junction is reverse biased. V_{CC} is of much higher value than V_{BB} . Fig. 18.16 shows the biasing arrangement for n-p-n transistor when the transistor has been represented by its symbolic form. Fig. 18.17 shows the same for a p-n-p transistor.

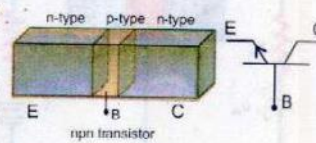


Fig. 18.14

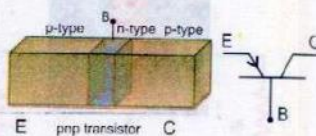


Fig. 18.15

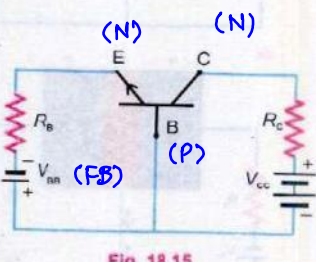


Fig. 18.16

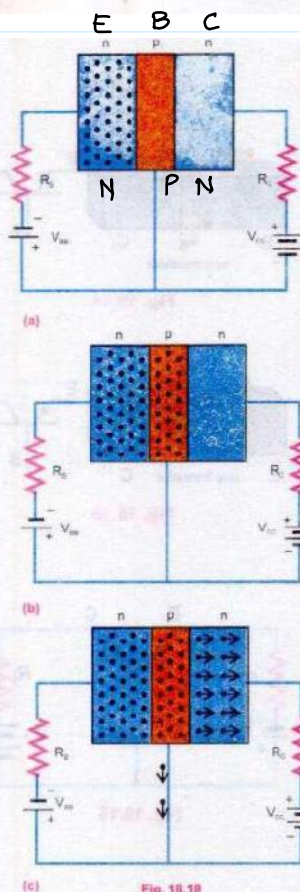


Fig. 18.18

Fig. 18.18 (a) shows a n-p-n transistor at the instant when the biasing voltage is applied. Electrons in the emitter, shown by black dots, have not yet entered the base region. After the application of the biasing voltage, emitter base junction is forward biased, so emitter injects a large number of electrons in base region (Fig. 18.18 b). These free electrons in the base can flow in either of two directions. They can either flow out of the base to the positive terminal of V_{BB} or they can be attracted towards the collector because of battery V_{CC} . Since the base is extremely thin, very few electrons manage to recombine with holes and escape out of the base. Almost all of the free electrons injected from the emitter into the base are attracted by the collector due to its large positive potential V_{CC} (Fig. 18.18 c). Thus, in a normally biased transistor due to above mentioned flow of electrons, we can say, that an electronic current I_E , flows from the emitter into the base. A very small part of it, current I_B , flows out of the base, the rest of it I_C flows out of the collector (Fig. 18.19)

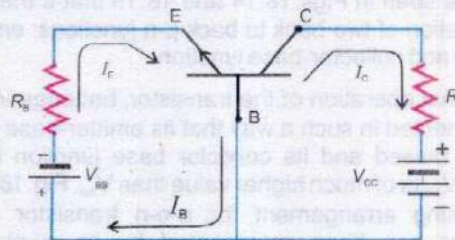


Fig. 18.19

The flow of conventional current is shown in Fig. 18.20. In future we will use conventional current only. From the figure it can be seen that

$$I_E = I_C + I_B \quad \dots \dots \dots (18.2)$$



$$I_E = I_C + I_B \quad \dots \dots \dots (18.2)$$

As very few electrons flow out of base, so I_B is very small a:

SIZE
 $C > E > B$

DOPING CONC
 $E > C > B$ # NPN

P TYPE $\rightarrow$ HOLES

N TYPE $\rightarrow e^-$

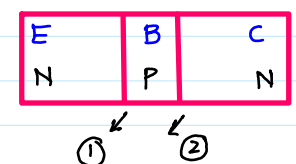
IVA + VA $\rightarrow a \rightarrow 1$
 $\rightarrow b \rightarrow 5$
 $\rightarrow c \rightarrow 10$

BJT $\Rightarrow$

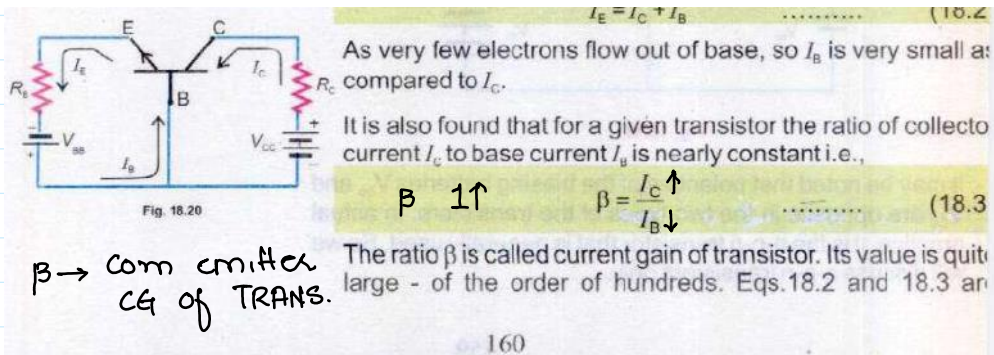
Bipolar Junction Transistor



PN JUNCTION



AMPLIF.



✓ **AMPLIF.**

x 20 times amplify

20 → $I_C \Rightarrow 20 I_B$

1 → I_B

fundamental equations of all transistors.

Example 18.1: In a certain circuit, the transistor has a collector current of 10 mA and a base current of 40 μ A. What is the current gain of the transistor? $\Rightarrow \beta$

Solution:

AMPLIF.

$$\beta = \frac{I_C}{I_B} = \frac{10 \times 10^{-3} \text{ A}}{40 \times 10^{-6} \text{ A}} = 250$$

$$\beta = \frac{I_C}{I_B} = \frac{I_C}{I_E - I_C} = \frac{\alpha I_E}{I_E - \alpha I_E}$$

$$\beta = \frac{\alpha I_E}{I_E(1 - \alpha)}$$

$$\alpha = \frac{I_C}{I_E} = \frac{I_C}{I_B + I_C} = \frac{\beta I_B}{I_B + \beta I_B} = \frac{\beta I_B}{I_B(1 + \beta)} = \frac{\beta}{1 + \beta}$$

$\alpha = \frac{I_C \downarrow}{I_E \uparrow} \rightarrow$ Comm. Base CG of Transistor.

$$\# \beta = \frac{I_C}{I_B} \Rightarrow I_C = \beta I_B$$

$$\# \alpha = \frac{I_C}{I_E} \Rightarrow I_C = \alpha I_E$$

$$\# I_E = I_B + I_C \rightarrow I_B = I_E - I_C$$

emitter-collector circuit.

Typically, Beta values range between 20 and 200 for most transistors. For instance, if a transistor has a Beta value of 100, one electron flows from the base terminal for every 100 electrons flowing between the emitter-collector terminals.

The mathematical relationship between Alpha (α) and Beta (β) expresses the current gain of the transistor. $\alpha = \frac{I_C}{I_E}$ and $\beta = \frac{I_C}{I_B} \therefore I_C = \alpha I_E = \beta I_B$

$$\text{As } \alpha = \frac{\beta}{\beta + 1} \text{ and } \beta = \frac{\alpha}{1 - \alpha}$$

$$I_E = I_C + I_B \dots \dots \dots 22.4$$

Where: " I_C " is the current flowing into the collector terminal, " I_B " is the current flowing into the base terminal and " I_E " is the current flowing out of the emitter terminal.

Worked Example 22.1

Find Alpha (α) rating of the transistor shown in diagram. Hence determine the value of I_C using both α and $\beta=49$ rating of the transistor.

Solution:

Step 1: Write down the known quantities and quantities to be found.

$$I_B = 240 \mu A, \quad I_E = 12 \text{ mA} \quad \text{and} \quad \beta = 49$$

$$I_C = ?$$

Step 2: Write down the formula and rearrange if necessary.

$$\alpha = \frac{\beta}{\beta + 1} = \frac{49}{49 + 1} = 0.98$$

The value of I_C can be found by using either α or β as under

$$I_C = \alpha \times I_E$$

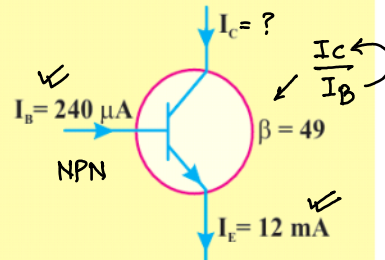
$$I_C = 0.98 \times 12 \text{ mA}$$

$$I_C = 11.76 \text{ mA}$$

$$I_C = \beta \times I_B$$

$$I_C = 49 \times 240 \mu A$$

$$I_C = 11.76 \text{ mA}$$



$$\# \beta = \frac{I_C}{I_B}$$

$$\# \alpha = \frac{I_C}{I_E}$$

Worked Example 22.2

In a common base connection, $I_C = 0.95 \text{ mA}$ and $I_B = 0.05 \text{ mA}$. Find the value of α .

$$\textcircled{1} I_E = I_C + I_B$$

$$\textcircled{2} \alpha = \frac{I_C}{I_E}$$

ANS: 0.95

18.7 OPERATIONAL AMPLIFIER

As stated earlier, amplifier is an important electronic circuit that is used in almost every electronic instrument. So instead of making amplifier circuit by discrete components, the whole amplifier is integrated on a small silicon chip and enclosed in a capsule. Pins connected with working terminals such as input, output and power supply project outside the capsule (Fig. 18.23 a). The enclosed circuit of the amplifier is used by making requisite connections with these pins. Such an

integrated amplifier is known as operational amplifier (op-amp), as it is some times used to perform mathematical operations electronically.

The op-amp is usually represented by its symbol shown in Fig. 18.23 (b). It has two input terminals. One is known as inverting input (-) and the other non-inverting input (+). A signal that is applied at the inverting (-) input, appears after amplification, at the output terminal with a phase shift of 180° (Fig. 18.24 a). It can be seen that the signal is inverted as it appears at the output. This is why this terminal is known as inverting. If the signal is applied at non-inverting input (+), it is amplified at the output without any change of phase (Fig. 18.24 b).

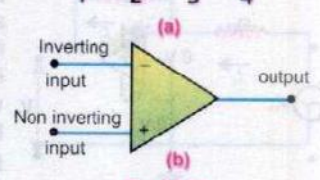
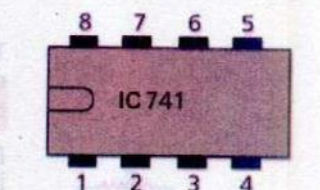
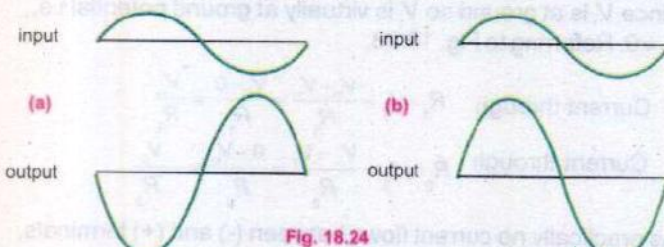
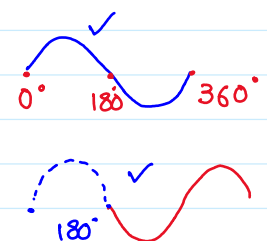
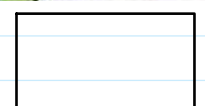


Fig. 18.23



BINARY SYSTEM

- ON - HIGH - (1) - TRUE
- OFF - LOW - (0) - FALSE

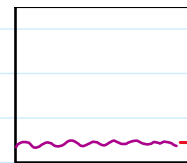


MOTOR $\Rightarrow$ OFF

BINARY SYSTEM

ON - HIGH - 1 - TRUE

OFF - LOW - 0 - FALSE



MOTOR $\Rightarrow$ OFF

MOTOR $\Rightarrow$ ON

SLIGHT $\rightarrow$ OFF
SLIGHT $\rightarrow$ ON

Voltage pulses
0 - 0.5V $\rightarrow$ OFF
2V - 5V $\rightarrow$ ON

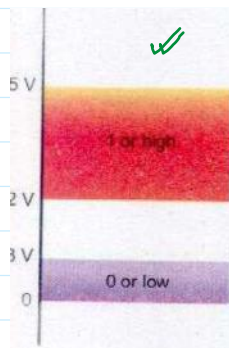
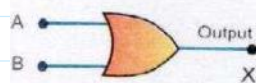


Fig. 18.34



operation. You have already read about these operations. Here we would study about logic gates which implement these operations.

18.13 FUNDAMENTAL LOGIC GATES

The electronic circuits which implement the various logic operations are known as logic gates. In these gates the high and low states, i.e., 1 and 0 states are simulated by certain voltage levels. Ideally one particular voltage level represents a high (1) and another voltage level represents a low (0). In practical digital circuits, however a 1 or high can be any voltage between a specified minimum value and a specified maximum value. Likewise 0 or low can be any voltage between a specified minimum and a specified maximum. Fig. 18.34 shows the range 1 and 0 levels for a certain type of digital gates. Thus if voltage of 3.5 V is applied to a gate, it will accept it as high or 1. If a voltage of 0.5 V is applied, the gate will recognize it as 0 or low.



Fig. 18.35

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

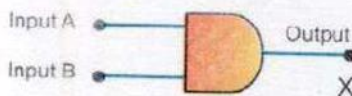


Fig. 18.36

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

will recognize it as 0 or low.

OR Gate

OR gate as symbolically represented in Fig. 18.35, implements the logic of OR operation. It has two or more inputs and a single output X. The output has a value 1 when at least one of its inputs A and B is at 1. Thus X will be zero only when both the inputs are 0. Thus it implements the truth table of OR operation (Table 18.2). The mathematical notation for OR operation is

$$X = A + B$$

- NED Test o/p ✓
- XII ✓

AND Gate $\Rightarrow$ اور $\Rightarrow$ saary ke saary

The AND gate shown in Fig. 18.36 has two or more inputs and a single output. It is designed such that it implements the truth table of AND operation, i.e., its output X is 1 only when both of its inputs A and B are at 1 and for all other combinations of the values of A and B, X is zero (Table 18.3). The mathematical notation for AND operation is

$$X = A \cdot B$$

$$X = A + B \Rightarrow 0 \times 0 = 0$$

$$X = A \cdot B$$

$$X = \bar{A}$$

NOT Gate

It performs the operation of inversion or complementation. That is why it is also known as inverter. It changes a logic level to its opposite level, i.e., it changes 1 to 0 and 0 to 1. The symbolic representation of NOT gate is shown in Fig. 18.37. Whenever a bar is placed on any variable, it shows that the value of the variable has been inverted. For example $\bar{1} = 0$ and $\bar{0} = 1$. The "bubble" (o) in Fig. 18.37 indicates operation of inversion. Its truth table is given in Table 18.4. The mathematical notation for NOT operation is $X = \bar{A}$

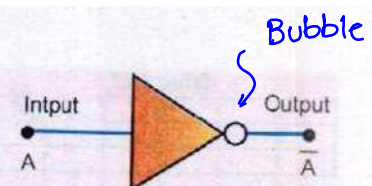


Fig. 18.37

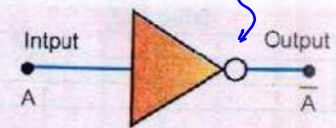
A	Output
0	1
1	0

18.14 OTHER LOGIC GATES

NOT Gate

$$X = \bar{A}$$

It performs the operation of inversion or complementation. That is why it is also known as inverter. It changes a logic level to its opposite level, i.e., it changes 1 to 0 and 0 to 1. The symbolic representation of NOT gate is shown in Fig. 18.37. Whenever a bar is placed on any variable, it shows that the value of the variable has been inverted. For example $\bar{1} = 0$ and $\bar{0} = 1$. The "bubble" (o) in Fig. 18.37 indicates operation of inversion. Its truth table is given in Table 18.4. The mathematical notation for NOT operation is $X = \bar{A}$.



NOT gate

Fig. 18.37

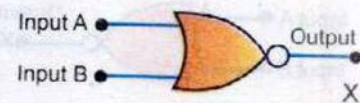
Table 18.4	
A	Output
0	1
1	0

18.14 OTHER LOGIC GATES

NOR Gate $\Rightarrow$ OR + NOT

In NOR gate the output of OR gate is inverted. Its symbol is shown in Fig. 18.38 and its truth table is given in Table 18.5. The mathematical notation for NOR operation is

$$X = \overline{A + B}$$



NOR gate

Fig. 18.38

Table 18.5		
A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

Fig. 18.38

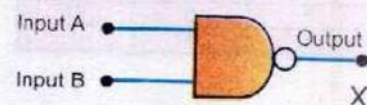
← NOR

NAND Gate

$$X = \overline{A \cdot B}$$

In NAND gate the output of an AND gate is inverted. Its symbol is shown in Fig. 18.39. The bubble in this figure shows that the output of AND gate is inverted. The truth table implemented by it is shown in Table 18.6. The mathematical notation for NAND operation is

$$X = \overline{A \cdot B}$$



NAND gate

Fig. 18.39

Table 18.6		
A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

Fig. 18.39

Exclusive OR Gate (XOR)

Consider a Boolean function X of two variables A and B such

A	B	AND	NAND
		$X = A \cdot B$	$X = \overline{A \cdot B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

Exclusive OR Gate(XOR)

Consider a Boolean function X of two variables A and B such that

$$X = A\bar{B} + \bar{A}B$$

The first term of the function X is obtained by ANDing the variable A with NOT of B . The second term is NOT of A ANDed with B . The function X is obtained by ORing these two terms. It can be constructed by combining AND, OR and NOT gates according to the scheme shown in Fig. 18.40(a). The

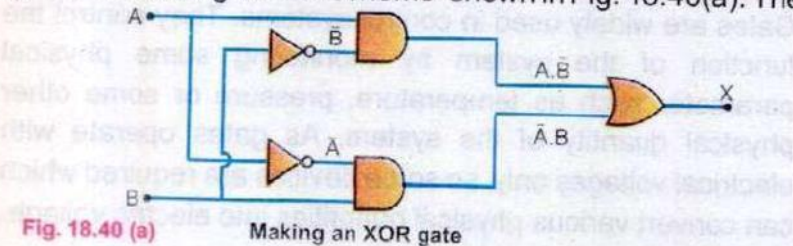
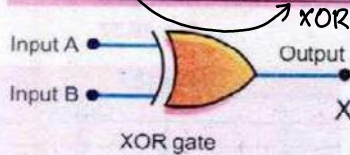


Fig. 18.40 (a)

Making an XOR gate

Table 18.7

A	B	Output
0	0	0
1	0	1
0	1	1
1	1	0



value of this function can be obtained by drawing the truth table (Table 18.7) which gives the value of X for all the values of the variables A and B . The value of X is 0 when the two inputs have the same values and it is 1 when the inputs have different values. It can be verified that the circuit of Fig. 18.40 (a) implements this truth table. The symbol of XOR gate is shown in Fig. 18.40(b).

Exclusive - NOR gate (XNOR)

A	B	$\bar{A}$	$\bar{B}$	$\bar{A}B$	$A\bar{B}$	$X = A\bar{B} \oplus \bar{A}B$
0	0	1	1	0	0	0
0	1	1	0	1	0	1
1	0	0	1	0	1	1
1	1	0	0	0	0	0



Fig. 18.40 (b)

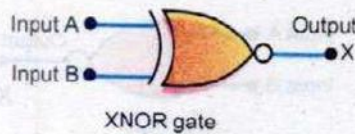


Fig. 18.41

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	1

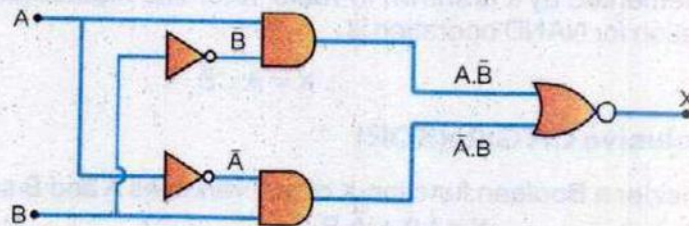
gate is shown in Fig. 18.40(b).

Exclusive - NOR gate (XNOR)

The exclusive NOR gate is obtained by inverting the output of a XOR gate. Its symbol is shown in Fig. 18.41. The bubble shown at the output in this figure shows that the output of XOR gate has been inverted. So its Boolean expression is given by

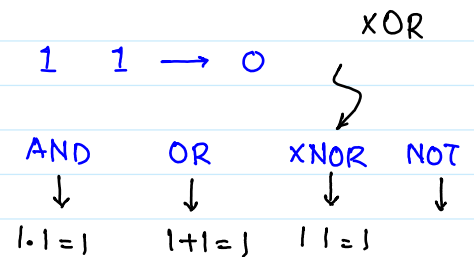
$$X = \overline{A\overline{B}} + \overline{\overline{A}B}$$

The truth table of XNOR gate is given in the Table 18.8. Its output is 1 when its two inputs are identical and 0 when the two inputs are different. Like XOR gate, it is also constructed by a combination of NOT, AND and NOR gates by the scheme shown in Fig.18.42.



The output of OR gate is 1

- (a) if both inputs are zero
- ☒ (b) if either or both inputs are 1
- (c) only if both input are 1
- (d) if either input is zero



If A and B are two inputs in AND gate, then AND gate has an output of 1 when the values of A and B are

- (a) A = 0, B = 0
- ☒ (b) A = 1, B = 1
- (c) A = 1, B = 0
- (d) A = 0, B = 1

A gate has the following truth table

P	Q	R
1	1	1
1	0	0
0	1	0
0	0	0

The gate is

- (a) NOR
- (b) OR
- (c) NAND
- ☒ (d) AND

The truth table given below is for

<i>A</i>	<i>B</i>	<i>Y</i>
0	0	1
0	1	1
1	0	1
1	1	0

(*A* and *B* are the inputs, *Y* is the output)

- (a) NOR (b) AND
(c) XOR (d) NAND

To use a transistor as an amplifier,

- (a) Both junctions are forward biased
(b) Both junctions are reverse biased
(c) the emitter-base junction is forward biased and the collector-base junction is reverse biased
(d) no biasing voltages are required

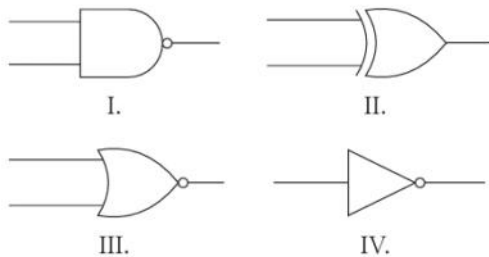
In the case of constants α and β of a transistor

- (a) $\alpha = \beta$ (b) $\beta < 1, \alpha > 1$ (c) $\alpha\beta = 1$ (d) $\beta > 1, \alpha < 1$

The base of transistor is made very thin compared to emitter and collector. By doing this, which one of the following is achieved?

- (a) High emitter current (b) Low power gain
(c) Low emitter current (d) High power gain

Given below are symbols for some logic gates.



The XOR gate and NOR gates respectively are

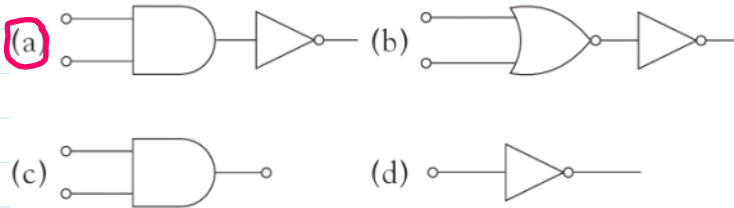
- (a) I and II (b) II and III
(c) III and IV (d) I and IV

The given symbol represents



- (a) NAND gate (b) OR gate
(c) AND gate (d) NOR gate

Which represents NAND gate?



Below we give four entries for the truth table of two point input OR gate. Which two are wrong?



	A	B	Y
i	1	0	1
ii	1	1	0
iii	0	0	1
iv	0	1	1

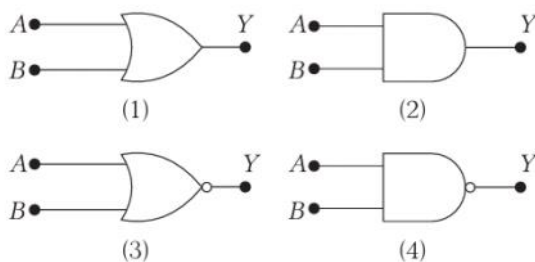
- (a) i, ii (b) ii, iii (c) iii, iv (d) iv, i

The below truth table corresponds to

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

- (a) NAND gate (b) XOR gate
(c) OR gate (d) NOR gate

Given below are four logic gate symbols (figure). Those for OR, NOR and NAND are respectively



- (a) 1, 4, 3 (b) 4, 1, 2 (c) 1, 3, 4 (d) 4, 2, 1

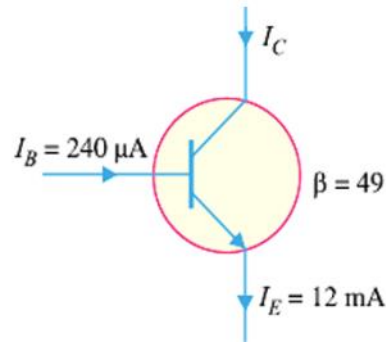
When the inputs of two input logic gates are 0 and 0, then the output is 1. When the inputs are 1 and 0, then the output is zero. The logic gate is of the type (a) XOR (b) NAND (c) NOR (d) OR

In a transistor, the value of β is 100, then the value of α will be

- (a) 0.01 (b) 0.1 (c) 0.99 (d) 1

$$\alpha = \frac{\beta}{\beta + 1}$$

5. In a common base connection, $I_E = 1\text{mA}$, $I_C = 0.95\text{mA}$. Calculate the value of I_B .
(0.05 mA)
6. Find the value of β if (i) $\alpha = 0.9$ (ii) $\alpha = 0.98$ (iii) $\alpha = 0.99$.
(9, 49, 99)
7. Calculate I_E in a transistor for which $\beta = 50$ and $I_B = 20\text{ }\mu\text{A}$.
(1.02 mA)
8. Find the α rating of the transistor shown in Fig.
Hence determine the value of I_C using both α and β rating of the transistor. (0.98, 11.76)



18.1 The current flowing into the base of a transistor is $100\text{ }\mu\text{A}$. Find its collector current I_C , its emitter current I_E and the ratio I_C/I_E , if the value of current gain β is 100. (Ans: 10 mA, 10.1 mA, 0.99)

Example 17.4

Calculate the emitter current in a transistor when its base current is $50\text{ }\mu\text{A}$ and the value of current gain is 100. **Ans: 5.05 mA**

How much is the emitter current with 100mA for the collector current and $800\text{ }\mu\text{A}$ for base current? (100.8mA)

A transistor has $\alpha = 0.995$. If the emitter current is 80mA , what is (i) the collector current (ii) the base current (iii) current gain β ?

(i) 79.6mA, (ii) 0.4mA, (iii) 199

The current gain β of a transistor is 200. If the base current is 0.2mA . Find (i) the collector current (ii) emitter current (iii) the value of α .

(i) 40mA (ii) 40.2mA (iii) 0.995